

Vpc peering

- VPC (Virtual Private Cloud) peering is a networking connection between two VPCs that enables them to communicate as if they were within the same network.
- It allows private communication between VPCs in the same region or across different regions, enabling you to route traffic using private IP addresses.
- VPC peering is commonly used in cloud environments like AWS, Google Cloud, and Azure, where it provides secure and cost-effective networking without the need for public internet traffic.

Created vpc-1

Your VPCs (1)

Info

Last updated
2 minutes ago

Actions

Create VPC

Search

vpc-1

Clear filters

<

1

>

☐	Name	VPC ID	State	IPv4 CIDR	IPv6 CIDR	DHCP c
☐	vpc-1	vpc-0074e7dd4bafdb4e8	Available	10.100.0.0/16	-	dopt-0:

Created igw and attached with vpc-1

[VPC](#) > [Internet gateways](#) > igw-0d1e228105443caac

igw-0d1e228105443caac / IGW-1

Actions ▼

Details [Info](#)

Internet gateway ID

 igw-0d1e228105443caac

State

 Attached

VPC ID

[vpc-0074e7dd4bafdb4e8](#) | [vpc-1](#)

Owner

 600627341010

Tags

Manage tags

 Search tags

< 1 > 

Key	Value
Name	IGW-1

Created two subnets within vpc-1

Subnets (2) [Info](#)

Last updated
10 minutes ago



Actions ▼

Create subnet

Find resources by attribute or tag

Name : PUB-SUB-1



Name : PRI-SUB-1



Clear filters

< 1 >



☐	Name	Subnet ID	State	VPC	IPv4 CIDR
☐	PUB-SUB-1	subnet-0ba84f26fdb15d9b5	✓ Available	vpc-0074e7dd4bafdb4e8 vpc-1	10.100.0.0/24
☐	PRI-SUB-1	subnet-086aabf0a979f151d	✓ Available	vpc-0074e7dd4bafdb4e8 vpc-1	10.100.1.0/24

Created two route tables using VPC-1

Route tables (6) Info							Last updated less than a minute ago		Actions ▼	Create route table
Find resources by attribute or tag							< 1 > 			
☐	Name ▼	Route table ID ▼	Explicit subnet associ... ▼	Edge associations ▼	Main ▼	VPC				
☐	2	rtb-00ca1c127857903c9	subnet-00cd0c51775feca...	–	No	vpc-07dcea9a7b6d1ae30				
☐	–	rtb-02866486ab672f8b6	–	–	Yes	vpc-02ca2daf68d136b4e				
☐	PUB-RT-1	rtb-08b853f4cb968f780	subnet-0ba84f26fdb15d...	–	No	vpc-0074e7dd4bafdb4e				
☐	–	rtb-0ba642b7d160cb430	–	–	Yes	vpc-07dcea9a7b6d1ae30				
☐	PRI-RT	rtb-07bb9e4345f25bcd0	subnet-086aabf0a979f15...	–	No	vpc-0074e7dd4bafdb4e				
☐	–	rtb-02749b52c88b40551	–	–	Yes	vpc-0074e7dd4bafdb4e				

Associated with subnet and updated the route

[VPC](#) > [Route tables](#) > rtb-08b853f4cb968f780

rtb-08b853f4cb968f780 / PUB-RT-1

Actions ▼

Details [Info](#)

Route table ID

 rtb-08b853f4cb968f780

Main

 No

Explicit subnet associations

[subnet-0ba84f26fdb15d9b5](#) / PUB-SUB-1

Edge associations

–

VPC

[vpc-0074e7dd4bafdb4e8](#) | [vpc-1](#)

Owner ID

 600627341010

[Routes](#)

[Subnet associations](#)

[Edge associations](#)

[Route propagation](#)

[Tags](#)

Explicit subnet associations (1)

Edit subnet associations

 Find subnet association

< 1 > 

Name



Subnet ID



IPv4 CIDR



IPv6 CIDR



PUB-SUB-1

[subnet-0ba84f26fdb15d9b5](#)

10.100.0.0/24

–

[Routes](#)

[Subnet associations](#)

[Edge associations](#)

[Route propagation](#)

[Tags](#)

Routes (2)

Both ▼

Edit routes

 Filter routes

< 1 > 

Destination



Target



Status



Propagated



0.0.0.0/0

[igw-0d1e228105443caac](#)

✔ Active

No

10.100.0.0/16

local

✔ Active

No

Launched two instance in vpc-1

Instances (2) [Info](#)

Last updated 2 minutes ago

Connect

Instance state ▼

Actions ▼

Launch instances ▼

All states ▼

Name = server-1 ✕

Name = server-2 ✕

Clear filters

< 1 >

☐	Name	Instance ID	Instance state ▼	Instance type ▼	Status check	Alarm status	Availability Zone ▼	Public IP v4
☐	server-2	i-00c652b06ef70c4bc	Running	t2.micro	2/2 checks passed View alarms +	us-east-1d	-	
☐	server-1	i-0a5f41f894287437c	Running	t2.micro	2/2 checks passed View alarms +	us-east-1d	-	

Select an instance

=

Created a new vpc and named it vpc-2

[VPC](#) > [Your VPCs](#) > vpc-07dcea9a7b6d1ae30

vpc-07dcea9a7b6d1ae30 / VPC-2

Actions ▼

Details [Info](#)

VPC ID  vpc-07dcea9a7b6d1ae30	State  Available	DNS hostnames Disabled	DNS resolution Enabled
Tenancy Default	DHCP option set dopt-03d3fe889f95bed76	Main route table rtb-0ba642b7d160cb430	Main network ACL acl-03327848e7277fd0f
Default VPC No	IPv4 CIDR 10.200.0.0/16	IPv6 pool -	IPv6 CIDR (Network border group) -
Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID  600627341010	

Created private subnet within vpc-2

VPC > Subnets > subnet-00cd0c51775feca3f

subnet-00cd0c51775feca3f / pri-sub-2

Actions ▼

Details

Subnet ID ☐ subnet-00cd0c51775feca3f	Subnet ARN ☐ arn:aws:ec2:us-east-1:600627341010:subnet/subnet-00cd0c51775feca3f	State ✔ Available	IPv4 CIDR ☐ 10.200.1.0/24
Available IPv4 addresses ☐ 250	IPv6 CIDR –	IPv6 CIDR association ID –	Availability Zone ☐ us-east-1f
Availability Zone ID ☐ use1-az5	Network border group ☐ us-east-1	VPC vpc-07dcea9a7b6d1ae30 VPC-2	Route table rtb-00ca1c127857903c9 2
Network ACL acl-03327848e7277fd0f	Default subnet No	Auto-assign public IPv4 address No	Auto-assign IPv6 address No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool –	Outpost ID –	IPv4 CIDR reservations –
IPv6 CIDR reservations –	IPv6-only No	Hostname type IP name	Resource name DNS A record Disabled
Resource name DNS AAAA record Disabled	DNS64 Disabled	Owner ☐ 600627341010	

Connection successfully established in a server-1(vpc-1)

```
ec2-user@ip-10-100-0-221:~  
Windows PowerShell  
Copyright (C) Microsoft Corporation. All rights reserved.  
  
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows  
  
PS C:\Users\kechiya> cd downloads  
PS C:\Users\kechiya\downloads> ssh -i "devops.08.pem" ec2-user@52.90.89.173  
  
A newer release of "Amazon Linux" is available.  
Version 2023.6.20241028:  
Version 2023.6.20241031:  
Run "/usr/bin/dnf check-release-update" for full release and version update info  
  
#_      _  
~\     #####_      Amazon Linux 2023  
~~   \#####\  
~~    \###|  
~~    \|/  ---> https://aws.amazon.com/linux/amazon-linux-2023  
~~     V~' '->  
~~~~  
~~ .-. /  
~-/_/_/  
~/m/'
```

```
Last login: Thu Nov 7 02:53:54 2024 from 103.190.145.35  
[ec2-user@ip-10-100-0-221 ~]$
```


A connection has been successfully established from server-1 to server-2(vpc-1)

```
[ec2-user@ip-10-100-0-120:~  
[ec2-user@ip-10-100-0-221 ~]$ ssh ec2-user@10.100.0.120  
ec2-user@10.100.0.120: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).  
[ec2-user@ip-10-100-0-221 ~]$ vi devops.pem  
[ec2-user@ip-10-100-0-221 ~]$ chmod 400 devops.pem  
[ec2-user@ip-10-100-0-221 ~]$ ssh ec2-user@10.100.0.120  
ec2-user@10.100.0.120: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).  
[ec2-user@ip-10-100-0-221 ~]$ ssh -i devops.pem ec2-user@10.100.0.120  
Last login: Thu Nov 7 03:03:03 2024 from 10.100.0.221
```

```
#_
~\_####_      Amazon Linux 2
~~\_#####\
~~~\###|
~~~~\#/
~~~~V~'--->
~~~~
~~~.._/
~~/_/_/_/_/_
~/m/'_/_/_/_/_
```

```
AL2 End of Life is 2025-06-30.
```

```
A newer version of Amazon Linux is available!
```

```
Amazon Linux 2023, GA and supported until 2028-03-15.  
https://aws.amazon.com/linux/amazon-linux-2023/
```

```
[ec2-user@ip-10-100-0-120 ~]$
```

To ensure seamless communication and enhanced security, we must implement vpc peering to establish secure connection between vpc-1 and vpc-2

- **Within the Same Cloud Provider:** VPC peering allows resources in separate VPCs to communicate privately, without using the public internet. For example, AWS VPC peering uses AWS's internal network, which is isolated from the public internet, adding a layer of security.
- **Cross-Account Connectivity:** It's often used to securely connect VPCs owned by different teams, departments, or business units, especially if they're in different accounts. This can be a more secure alternative to internet or VPN connections.

[VPC](#) > [Peering connections](#) > pcx-0159286e9ba723acb

pcx-0159286e9ba723acb / vpc peering 1-2

Actions ▼

Details Info

Requester owner ID
 600627341010

Peering connection ID
 pcx-0159286e9ba723acb

Status
 Active

Expiration time
—

Accepter owner ID
 600627341010

Requester VPC
[vpc-0074e7dd4bafdb4e8 / vpc-1](#)

Requester CIDRs
 10.100.0.0/16

Requester Region
N. Virginia (us-east-1)

VPC Peering connection ARN
 arn:aws:ec2:us-east-1:600627341010:vpc-peering-connection/pcx-0159286e9ba723acb

Accepter VPC
[vpc-07dcea9a7b6d1ae30 / VPC-2](#)

Accepter CIDRs
 10.200.0.0/16

Accepter Region
N. Virginia (us-east-1)

Update the routes to vpc peering

[VPC](#) > [Route tables](#) > [rtb-00ca1c127857903c9](#) > Edit routes

Edit routes

Destination	Target	Status	Propagated
10.200.0.0/16	local	Active	No
	Q local		
Q 10.100.1.0/24	Peering Connection	Active	No
	Q pcx-0159286e9ba723acb		
Add route			

Updating inbound rules of instance-1(sg)

[VPC](#) > [Security Groups](#) > [sg-073cd9490915f75f3 - PRI-SG](#) > Edit inbound rules

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-0e9dec8bfae7c78d9	All ICMP - IPv4 ▼	ICMP	All	Custom ▼	Q 10.100.1.0/24 ✕	Delete
sgr-0bb352e88b6885cf9	SSH ▼	TCP	22	Custom ▼	Q 10.100.1.0/24 ✕	Delete

Add rule

Cancel

Preview changes

Save rules

Updated route in pri-rt(vpc-1)

VPC > Route tables > rtb-07bb9e4345f25bcd0 > Edit routes

Edit routes

Destination	Target	Status	Propagated
10.100.0.0/16	local	Active	No
Q 10.200.1.0/24	local		
	Peering Connection	Active	No
	Q pcx-0159286e9ba723acb		

